

ENGINEERING CASE STUDY & DEVOPS SHOWCASE

Workplace Assessment & Proctoring Platform

A production-grade, zero-trust examination system engineered with browser-native video proctoring, server-authoritative scoring, multi-stage Docker containerization, and full Prometheus & Grafana observability.

 Multi-Stage Docker

 Prometheus Telemetry

 Grafana Dashboards

 Railway Cloud Deployment

 GitHub Actions CI/CD

 WebRTC & MediaRecorder

 WebSockets / Socket.io

 TypeScript & React 19

 Prisma ORM & SQLite

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[View Live Portfolio](#) ↗

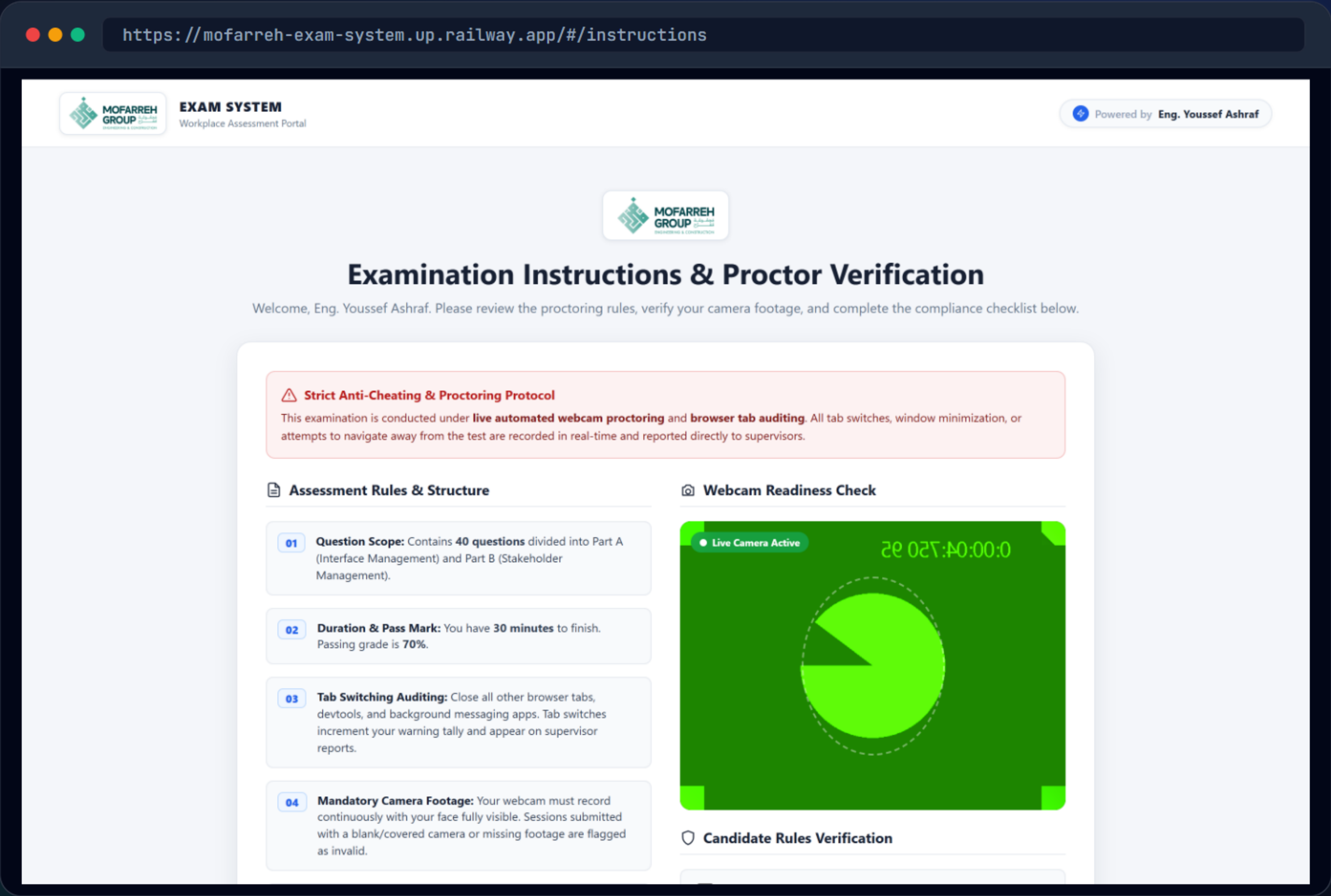
Available for Hire in 2 Months (Post-Military)

Swipe through for Architecture, Live UX & Observability →

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Browser–Native Camera Proctoring & Integrity Rules

Zero third-party proprietary plugins. Built entirely on modern HTML5, MediaStream, and Canvas Web APIs.



🖥️ Face–Alignment Guide

Interactive SVG oval overlay guarantees optimal lighting, centered framing, and live camera feed verification before test entry.

💾 IndexedDB Chunk Buffering

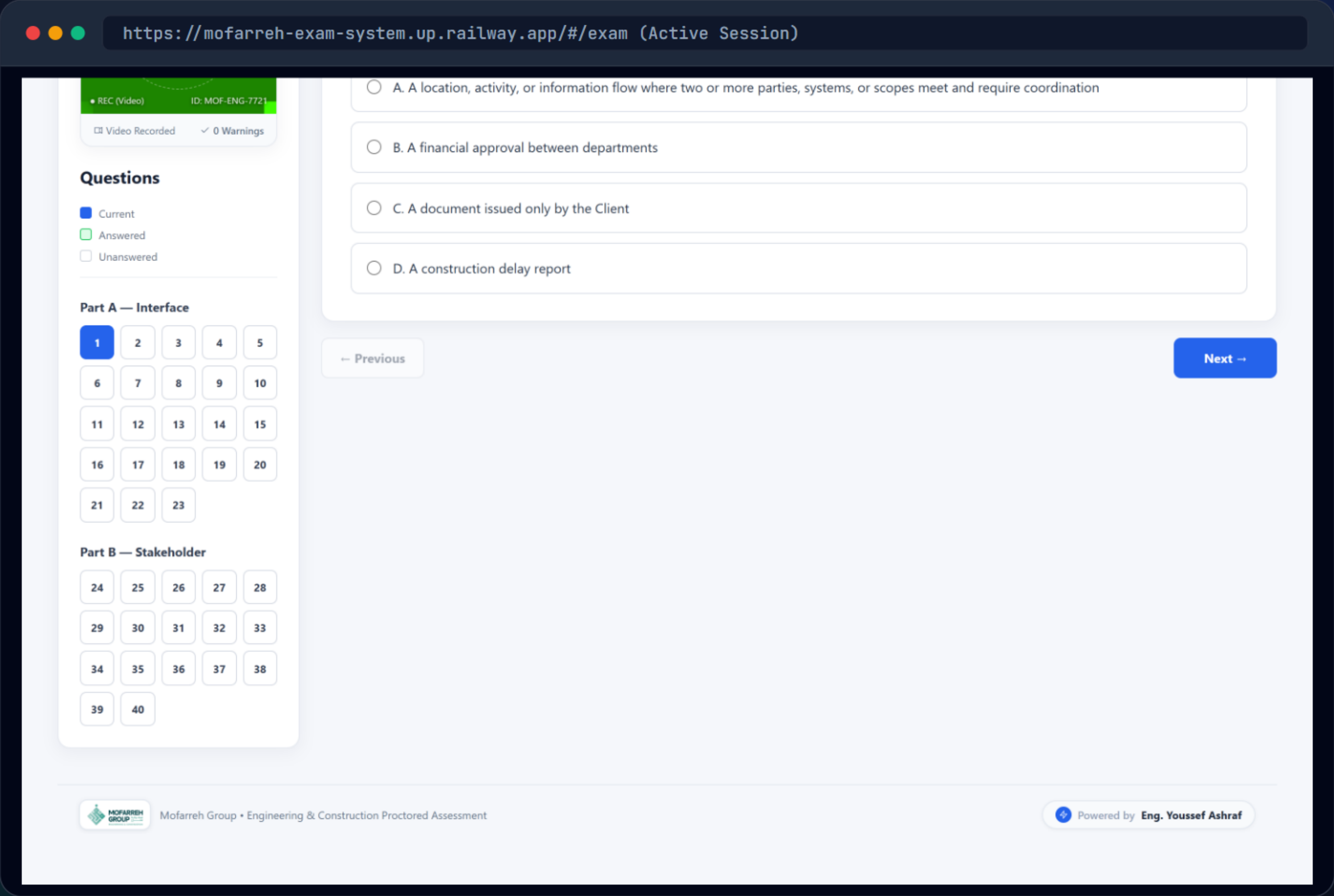
Records the entire sitting in time slices directly into client–side IndexedDB, preventing memory bloat before secure POST upload.

🔔 Window Blur Audit

Tab switching triggers candidate:warning events banking server–side warnings and locking attempts on excessive violations.

Live Timed 40–Question Examination Sitting

Two-part competency assessment with live recording indicators and synchronized question navigation.



● Live Recording Badge

Active • REC (Video) indicator reassures compliance while MediaRecorder streams silent chunk uploads in the background.

🕒 Absolute Deadline Clock

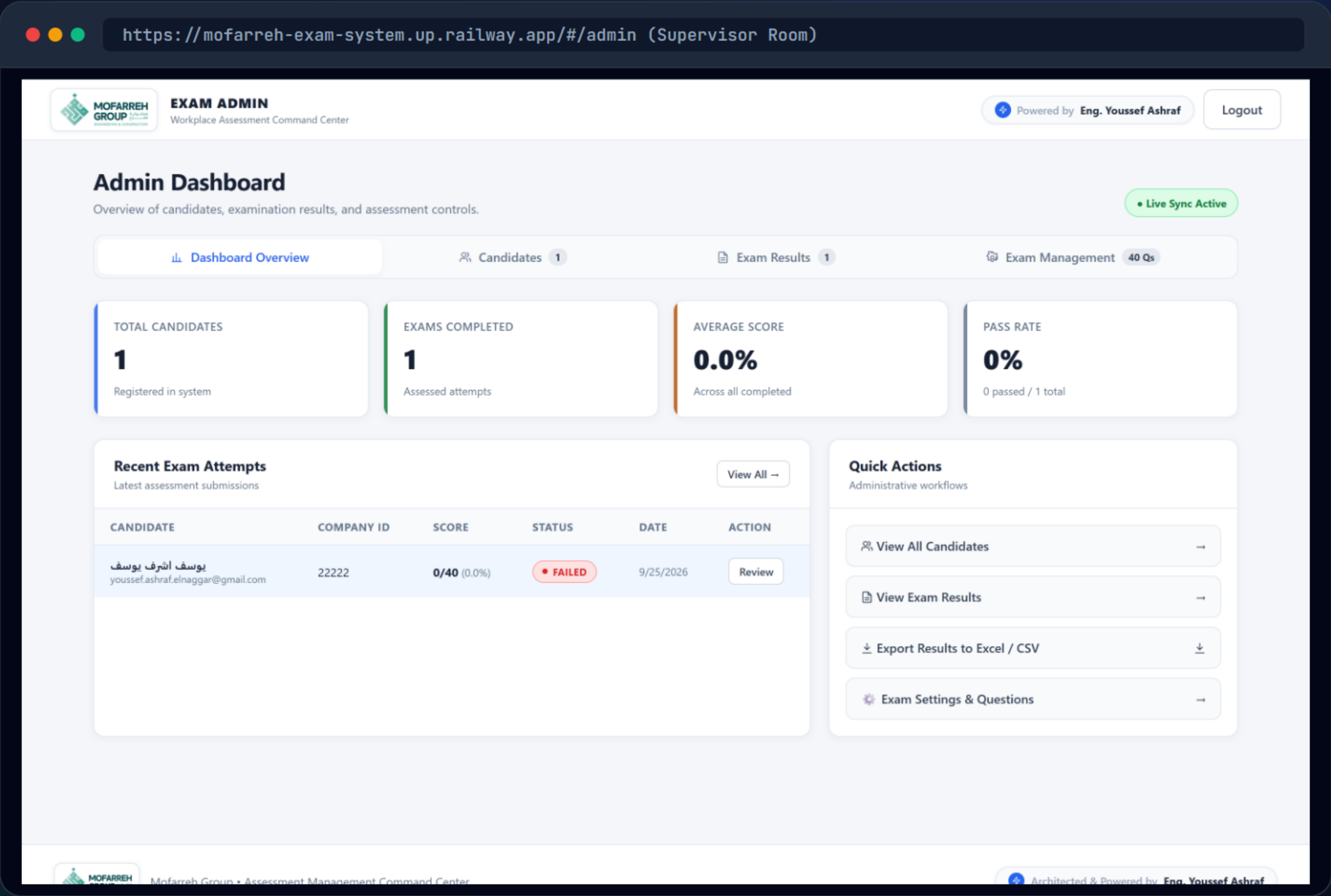
Timestamp-based expiration prevents timer manipulation from page refreshes, tab throttling, or network drops.

📁 Split Matrix Layout

Interactive 40–question grid showing answered, flagged, and active questions across Part A & Part B modules.

Supervisor Command Center & Real-Time Sync

Instant candidate tracking, server-authoritative scoring, and transactional supervisor notifications.



 **Server-Owned Scoring**

Correct answers are omitted from client bundles. Scoring, attempt counters, and 48-hr lockouts are purely server-enforced.

 **Real-Time Push Events**

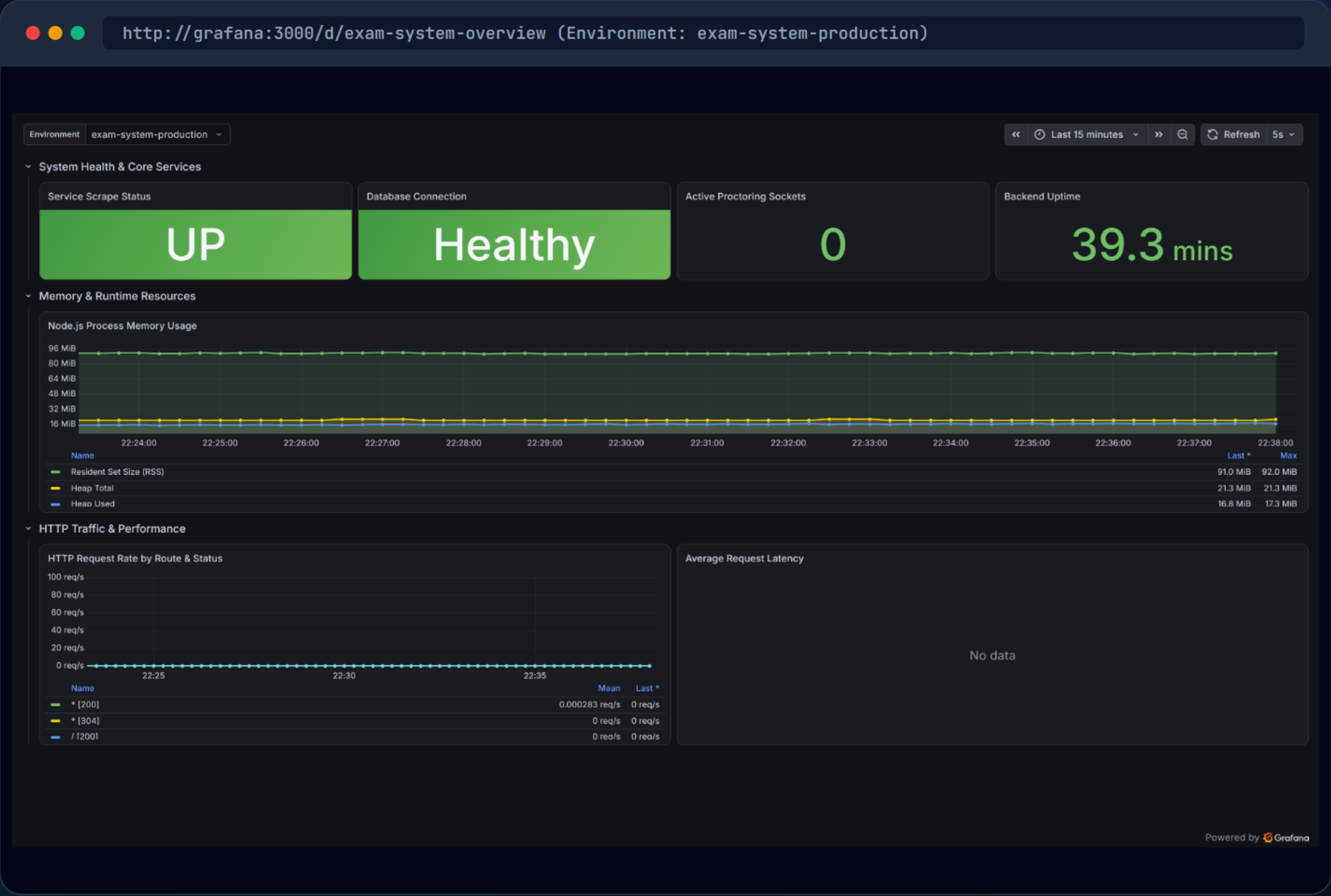
Socket.io rooms synchronize active examinees, live warnings, and completion timestamps instantly to all connected supervisors.

 **Resend HTTPS Delivery**

Transactional supervisor alerts dispatching score cards and proctoring summaries reliably over outbound HTTPS port 443.

Production Observability with Grafana

Continuous monitoring of container health, Node.js memory profiling, and HTTP request throughput.



Core Health Status

Instant visibility into live Scrape Status (UP), Prisma/SQLite connectivity (Healthy), and continuous backend uptime.

Memory Profiling

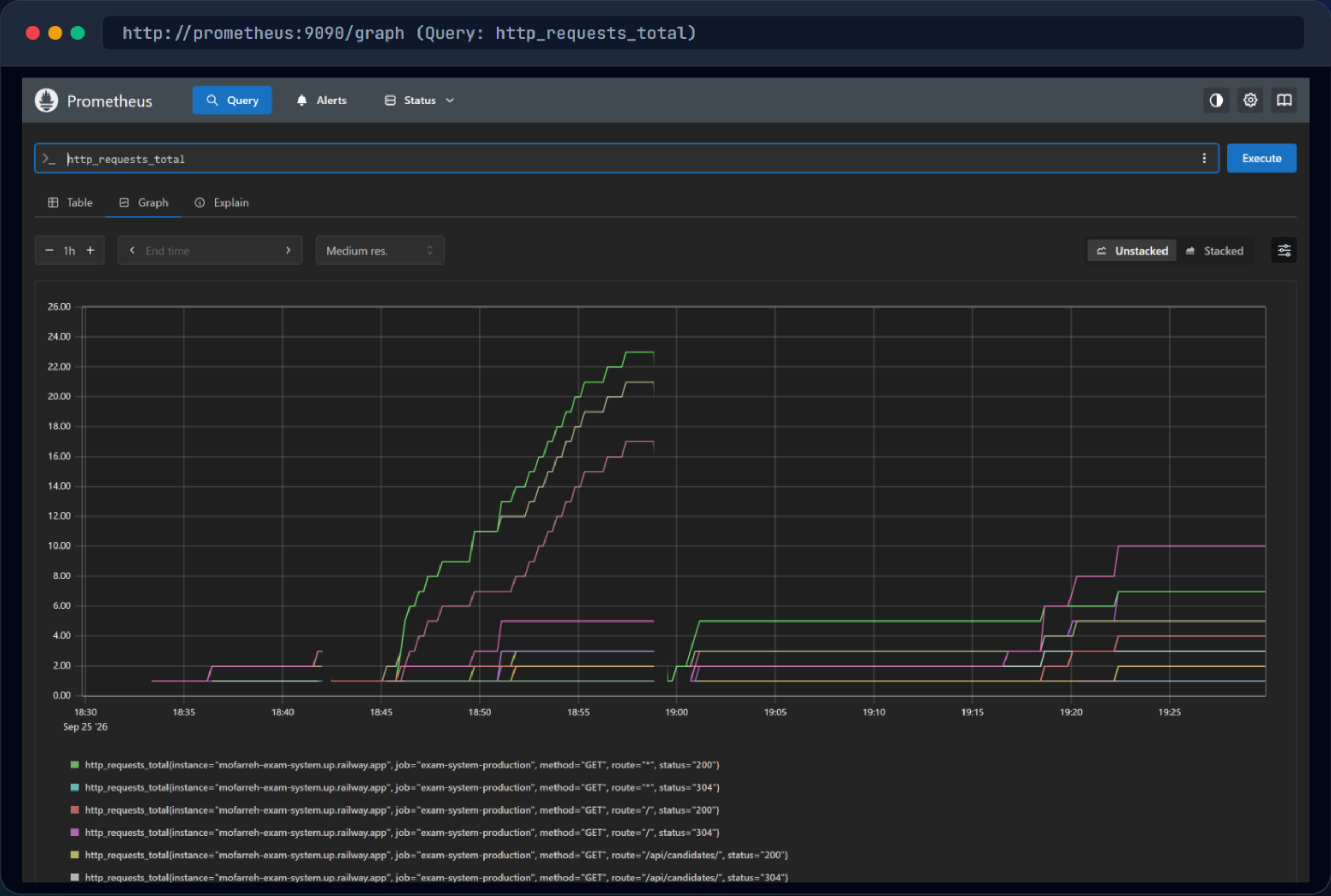
Real-time metrics tracking Resident Set Size (~92 MiB), Heap Allocated (21.3 MiB), and Heap Used (16.8 MiB) under load.

HTTP Request Rates

Per-route request breakdown graphing latency distributions and traffic status codes (200 OK, 304 Not Modified).

Prometheus Exporter & Time-Series Engine

Custom Prometheus metrics endpoint scraping cloud production targets with sub-300ms latency.



Custom Prometheus Metrics

Custom Prom metrics (`/metrics`) reporting database latency, active WebSockets, memory, and HTTP traffic.

Sub-300ms Scrape Latency

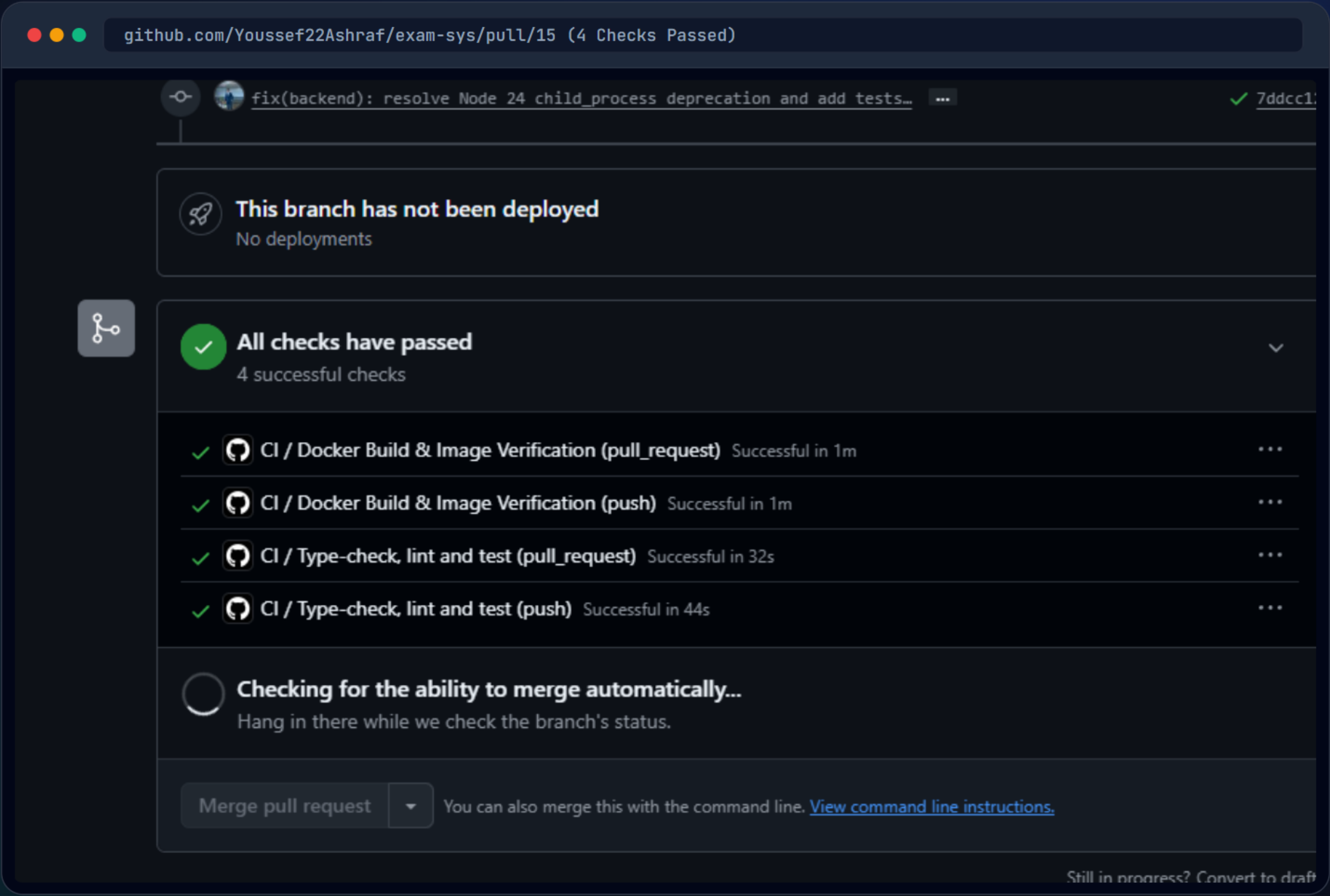
Lightweight exporter returns metrics in ~273ms without blocking the Node.js event loop or taxing container CPU.

PromQL Time-Series

Full query engine evaluating `rate(http_requests_total[5m])`, failure spikes, and proctoring socket counts.

Automated CI/CD Quality Gates

GitHub Actions pipeline enforcing TypeScript compilation, ESLint, unit tests, and Docker builds.



Docker Build Verification

Builds and validates production Docker container image on pull request before merge to guarantee build stability.

TypeScript & ESLint

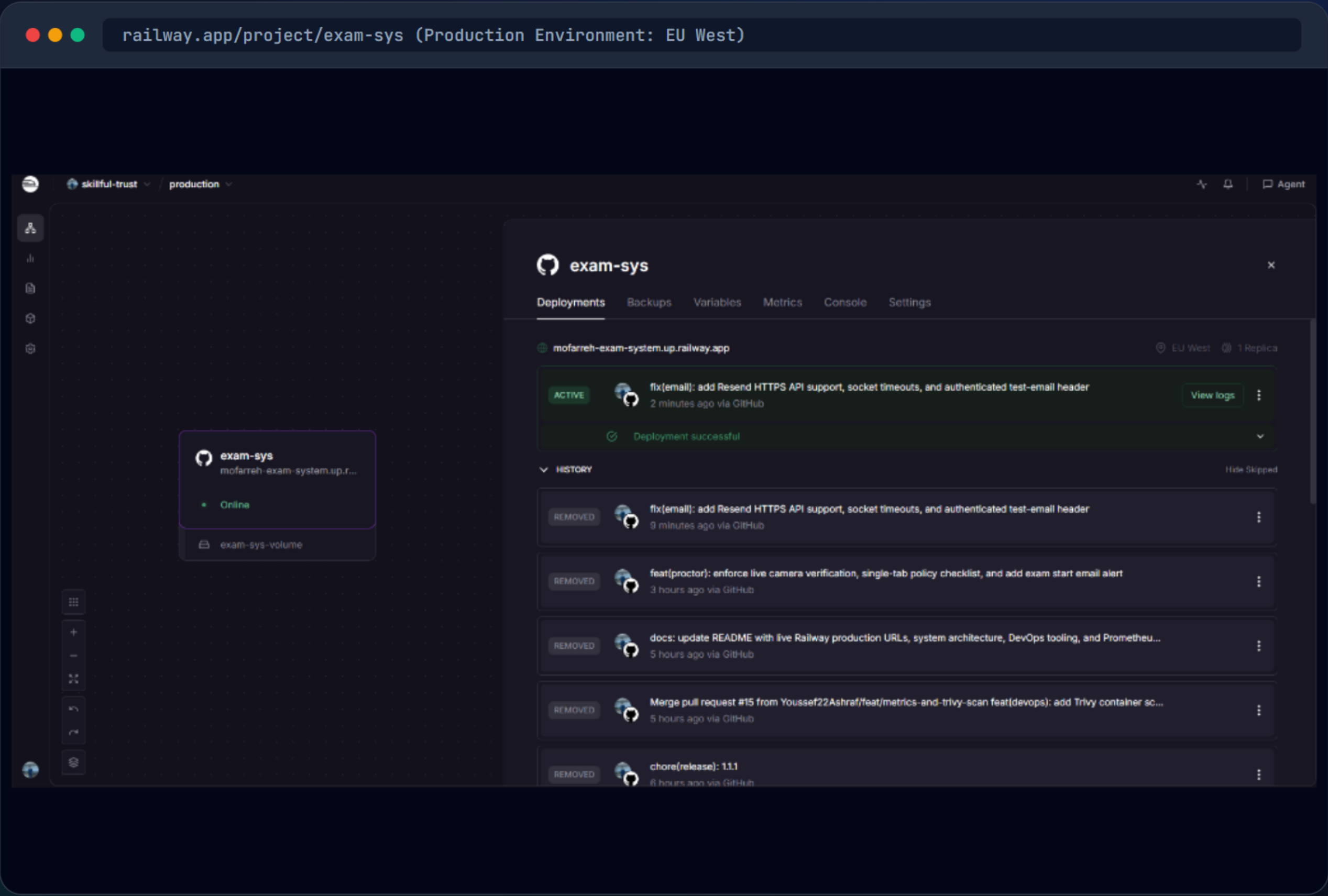
Zero-compromise static analysis: tsc -b frontend and tsc backend run with strict flags to prevent regressions.

Automated Test Suite

Automated unit and integration testing of pure scoring logic, cooldown algorithms, and route endpoints.

Cloud Production Deployment on Railway

Multi-stage Docker container deployed with persistent volumes and edge SSL routing.



🚀 Unified Single-Port Hosting

Express API + Vite SPA bundled into one container on port 5000, eliminating CORS and proxy complexity.

💾 Persistent Volume Mounts

Mounted volumes (/app/prisma, /app/uploads) ensure database state and video archives persist across re-deploys.

🔄 Git-Triggered Deployments

Automated zero-downtime rolling updates on push to main branch with automated migrations and health checks.